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SIMATS ENGINEERING
ASSESSMENT TOOLS

COURSE NAME: **Theory of Computation** **COURSE CODE:** **CSA13**

COURSE OUTCOME COVERED

CO3: Construct regular expressions, and use the pumping lemma and closure properties to analyze regular languages. (BL:3)

Assessment Tool Weightage: 40%

QUESTIONS: 5 **Total Marks:25**
Duration : 1 Hour

Experiments

Code of Conduct:

I A·Manoj Reddy 19231171 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Manoj

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ASSESSMENT TOOLS

1 Write a C program to check whether a given string belongs to the language defined by a Context Free Grammar (CFG)

$S \rightarrow 0A1 \qquad A \rightarrow 0A \mid 1A \mid \varepsilon$

2 Write a C program to check whether a given string belongs to the language defined by a Context Free Grammar (CFG)

$S \rightarrow 0S0 \mid 1S1 \mid 0 \mid 1 \mid \varepsilon$

3 Write a C program to check whether a given string belongs to the language defined by a Context Free Grammar (CFG)

$S \rightarrow 0S0 \mid A \qquad A \rightarrow 1A \mid \varepsilon$

4 Write a C program to check whether a given string belongs to the language defined by a Context Free Grammar (CFG)

$S \rightarrow 0S1 \mid \varepsilon$

5 Write a C program to check whether a given string belongs to the language defined by a Context Free Grammar (CFG)

$S \rightarrow A101A \qquad A \rightarrow 0A \mid 1A \mid \varepsilon$

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RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	10	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

86/100

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main.c

Output

Question 1

```c
#include <stdio.h>
#include <string.h>

int main() {
    char s[100];
    int i, n;

    printf("Enter string: ");
    scanf("%99s", s);
    n = strlen(s);

    if (n >= 2 && s[0] == '0' && s[n-1] == '1') {
        for (i = 1; i < n-1; i++)
            if (s[i] != '0' && s[i] != '1')
                break;

        if (i == n-1)
            printf("String belongs to the CFG.");
        else
            printf("String does not belong to the CFG.");
    } else {
        printf("String does not belong to the CFG.");
    }

    return 0;
}
```

Output

```
Enter string: 00101
String belongs to the CFG.
```

CFG: $S \rightarrow 0A1$ $A \rightarrow 0A \mid 1A \mid \varepsilon$

Compiled and executed successfully

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main.c Output

Question 2

```c
#include <stdio.h>
#include <string.h>

int main() {
    char s[100];
    int i, n, flag = 1;

    printf("Enter string: ");
    scanf("%99s", s);
    n = strlen(s);

    for (i = 0; i < n / 2; i++) {
        if (s[i] != s[n - 1 - i]) {
            flag = 0;
            break;
        }
    }

    if (flag)
        printf("String belongs to the CFG.");
    else
        printf("String does not belong to the CFG.");

    return 0;
}
```

Output

```
Enter string: 0110
String belongs to the CFG.
```

CFG: $S \rightarrow 0S0 \mid 1S1 \mid 0 \mid 1 \mid \varepsilon$

Compiled and executed successfully

main.c Output

Question 3

```c
#include <stdio.h>
#include <string.h>

int main() {
    char s[100];
    int i, n, left = 0, right, flag = 1;

    printf("Enter string: ");
    scanf("%99s", s);
    n = strlen(s);
    right = n - 1;

    while (left < n && s[left] == '0')
        left++;

    while (right >= 0 && s[right] == '0')
        right--;

    for (i = left; i <= right; i++) {
        if (s[i] != '1') {
            flag = 0;
            break;
        }
    }

    if (flag && left == n - 1 - right)
        printf("String belongs to the CFG.");
    else
        printf("String does not belong to the CFG.");

    return 0;
}
```

Output

```
Enter string: 0011100
String belongs to the CFG.
```

CFG: $S \rightarrow 0S0 \mid A \quad A \rightarrow 1A \mid \varepsilon$

Compiled and executed successfully

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main.c Output

Question 4

```c
#include <stdio.h>
#include <string.h>

int main() {
    char s[100];
    int i, n, zeros = 0, ones = 0, flag = 1;

    printf("Enter string: ");
    scanf("%99s", s);
    n = strlen(s);

    for (i = 0; i < n; i++) {
        if (s[i] == '0')
            zeros++;
        else if (s[i] == '1')
            ones++;
        else {
            flag = 0;
            break;
        }

        if (i > 0 && s[i] == '0' && s[i-1] == '1')
            flag = 0;
    }

    if (flag && zeros == ones) {
        for (i = 0; i < zeros; i++) {
            if (s[i] != '0' || s[n-1-i] != '1')
                flag = 0;
        }
    }

    if (flag)
        printf("String belongs to the CFG.");
```

Output